Is Maia-2 too sure of itself?

A temperature for each rating band, fitted on Lichess and tested on the player's games

Chess Coach research lab

Abstract

The coaching app reads Maia-2's probabilities as chances: the odds that a mistake is punished, how findable a move is, the moves its Maia bot samples. The lab's benchmark found Maia-2 more confident than right on Lichess in every band. We fitted a temperature for each band on 12,474 Lichess moves and tested it by a rule fixed before any number was computed. It improves the held-out Lichess moves, but it makes the player's own games less likely and the punish odds worse, so it is not adopted. On his games, at the rating offsets the app uses, Maia-2 is already about as sure as it is right, and softening it makes it less sure than right. The app keeps the raw policy. Scored by the app's own Maia-2, the Lichess sample also reproduces the benchmark's first-choice count exactly.

1 The question

The coaching app reads Maia-2's probabilities as chances. A review says how often players at the opponent's level find the reply that punishes a mistake; a drill says how findable a move is at the player's level; the Maia bot samples its moves from them. The lab's benchmark of Maia-2, on 25,000 moves from rated rapid games on Lichess in June 2025, after Maia-2's training window, found it more confident than right in every band: its first choice is the move played 51.5% of the time, while the mean probability it gives that choice is 54.9%, and the gap runs from 3.0 to 3.9 points across the benchmark's bands.

A common remedy is a temperature: each probability p becomes $p^{1/T}$, divided by the same sum over the position's legal moves. A temperature above one softens the policy and below one sharpens it; either way the order of the moves stays, so the top move and every ranking the app reads stay as they are. Does a temperature for each band, fitted on Lichess, make the probabilities truer on the player's own games?

2 The data

Fitting. The benchmark's own sample, drawn again by its sampler, unchanged: 5,000 moves per band of the benchmark from 150,181 streamed games, at most 8 per player and game, from ply 11 on with both clocks at 30 seconds or more. Each move is scored by the Maia-2 the app ships, through its engine service in Node, with each side at its own Lichess rating's band. The 13,785 games, ordered by their ids, alternate between a fitting half (12,474 moves) and a held-out half (12,526 moves).

Testing. The player's games, never used to fit: the 36,702 moves of the rating-offset project in 697 games, with each player's chess.com rating converted at the app's offset for that player: 250 for him and 400 for his opponents (the rating-offset project's single offset for him, the player-offsets project's for them); and the punish events of the punish-odds project, each priced as the app prices it, at the replier's band, so converted, facing an equal opponent. The punish events' primary set holds the 579 replies whose evidence the review's deeper pass verified; all 679 non-book replies, in 60 games, are reported beside it.

Parity. Scored by the app's Maia-2, the move played is its first choice for 12,875 of the 25,000 moves, exactly the count the benchmark's Maia-2, the reference implementation, gives on the same moves.

3 The candidate and the rule

Each band's temperature maximises the likelihood of the moves played by movers in that band in the fitting half; a band with fewer than 300 fitting moves would keep $T = 1$. The rule was committed before any number was computed: the temperatures are adopted only if all three tests hold, each judged by the 95% interval of the difference over 1,000 bootstrap draws of games.

1. On the Lichess held-out half, the mean log-likelihood per move of the moves played rises.
2. On the player's games, the same.
3. On the punish events, the Brier score of the punish odds, against whether the punishing reply was played, does not rise.

4 Results

Not adopted: the temperatures pass the first test and fail the second and the third (Table 1). Fitted on Lichess, every band's temperature is above one, from 1.02 to 1.13 (Figure 1), so every band is softened; each had at least 620 fitting moves. On the Lichess held-out half, the softer policy raises the mean log-likelihood of the moves played by 0.0029 per move [0.0014, 0.0044]. On the player's games it lowers it by 0.0054 per move [0.0043, 0.0064], and the rule stops there. The punish events fail too: their Brier score rises from 0.164 to 0.167, by 0.0024 [0.0002, 0.0048], an interval entirely above zero. On all non-book replies the rise, 0.0010 [−0.0010, 0.0031], holds zero.

test	raw	tempered	tempered − raw	
1. Lichess, held-out half (log-likelihood per move)	−1.4944	−1.4915	0.0029 [0.0014, 0.0044]	passes
2. the player's games (log-likelihood per move)	−1.3883	−1.3936	−0.0054 [−0.0064, −0.0043]	fails
3. punish events, verified (Brier score)	0.1641	0.1665	0.0024 [0.0002, 0.0048]	fails
punish events, all (Brier score)	0.1526	0.1536	0.0010 [−0.0010, 0.0031]	

Table 1: The three tests, raw against tempered, with the 95% interval of the difference over 1,000 bootstrap draws of games. A higher log-likelihood and a lower Brier score are better. All non-book punish events are reported beside the verified ones and decide nothing.

Why it does not transfer. Figure 2 and Table 2 compare Maia-2's mean probability of its first choice with how often that choice is the move played. On the Lichess held-out half Maia-2 is more

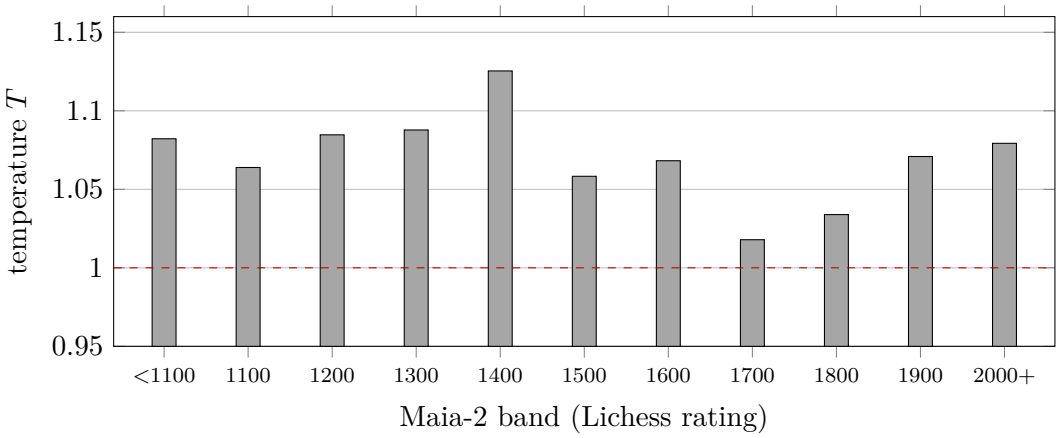


Figure 1: Each band's temperature, fitted on the Lichess fitting half. Above the dashed line a band's policy is softened.

confident than right in every band, by 2.0 to 5.2 points, and the temperature brings that to between −1.1 and 3.4. On the player's games there is less to correct: in the bands he and his opponents occupy (<1100 to 1600), the gap runs from −0.7 to 2.7 points, and over all his games' moves it is 0.8 points against 3.4 on the Lichess held-out half. Softening turns every band of his games but one underconfident, by up to 3.1 points; in the 1500 band it lands 0.04 points above how often the first choice is played. One reading, not tested here: the rating offsets were chosen on these games for the same likelihood, and a higher band is a sharper policy, so the offsets already set how sure Maia-2 is about them.

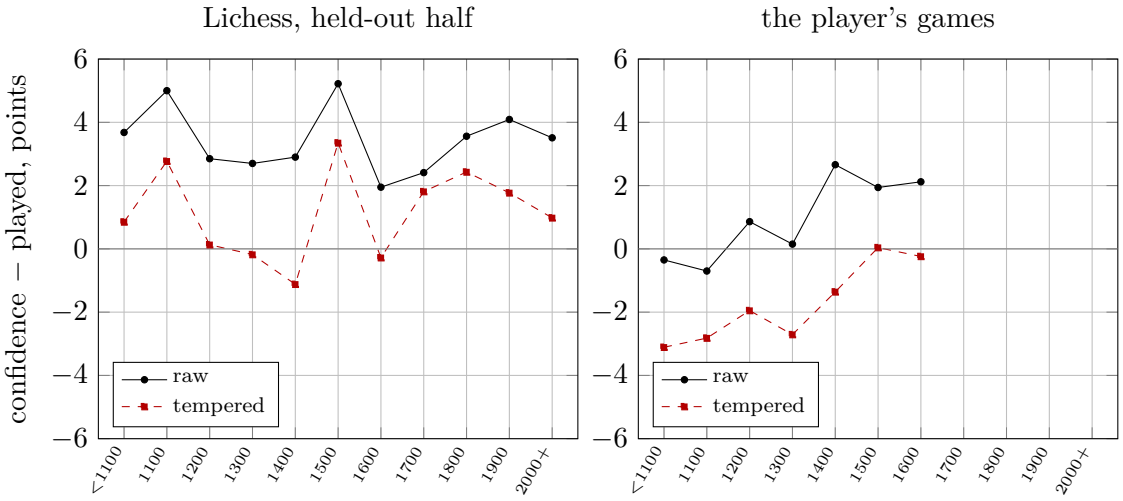


Figure 2: Maia-2's mean probability of its first choice less the share of moves where that choice was played, in points, by band: above zero it is more confident than right.

			Lichess, held-out half			the player's games			
band	fitting moves	T	played	raw	tempered	moves	played	raw	tempered
<1100	2,537	1.082	46.2	49.8	47.0	4,098	52.6	52.2	49.5
1100	729	1.064	49.2	54.2	52.0	4,904	55.0	54.3	52.2
1200	833	1.085	51.3	54.2	51.5	5,782	53.6	54.5	51.7
1300	946	1.088	52.2	54.9	52.0	10,948	54.9	55.1	52.2
1400	814	1.125	53.1	56.0	52.0	6,367	53.1	55.8	51.7
1500	823	1.058	49.2	54.4	52.5	4,580	54.5	56.5	54.6
1600	841	1.068	52.8	54.7	52.5	23	47.8	49.9	47.6
1700	1,001	1.018	55.2	57.6	57.0				
1800	833	1.034	53.4	56.9	55.8				
1900	620	1.071	53.3	57.4	55.1				
2000+	2,497	1.079	54.2	57.7	55.2				

Table 2: By band: the fitting moves and the fitted temperature, then, on the Lichess held-out half and on the player's games, the share of moves where Maia-2's first choice was played and its mean probability, raw and tempered (percent).

5 What it means for the app

Nothing to port. On this player's games Maia-2, at the bands the app's offsets give, is about as sure as it is right, so the punish odds, findability and the Maia bot keep the raw policy. The overconfidence the benchmark measured is real on Lichess; it is not what this player's games show.

6 Limitations

- One temperature per band corrects confidence on average. It cannot fix a policy that ranks the moves wrongly, and it says nothing about positions where Maia-2 is sure for a reason.
- The fitting sample is one month of Lichess rapid. The player's games are chess.com rapid, through a rating conversion that is itself fitted.